HandoffHQ — Architecture & Implementation Blueprint

0. Understanding, Verified Baseline & Explicit Assumptions

Confirmed Product Baseline

- **Hackathon & Track:** Kalvium Hackathon — Track 3 (Notion / CRM / Agency Operations).
- **Product Definition: HandoffHQ** is a **Sales-to-Delivery Handover Command Center** for service agencies. It manages the transition from a newly won client deal to an execution-ready project for the delivery team.
- **Winning Differentiator:** A deterministic, rule-based **Readiness Engine** that evaluates mandatory handover requirements in real time, computes completion percentage and status (Draft, Blocked, Ready for Delivery, Handed Over), explains *why* each missing requirement blocks the transition, provides actionable remediation steps, prevents premature handoffs, and logs an immutable activity audit trail.
- **Anchor Judge Demo:** A 60-second interactive journey centered on the **"Acme Website Redesign"** handover, demonstrating a blocked handover, live blocker resolution, role switching from Sales Rep to Delivery Lead, final acceptance, and persisted audit history.

Explicit Technical & Product Assumptions

Because no further questions are needed that would materially alter the architecture, the following concrete assumptions are locked for the MVP:

1. **Assumption A-01 (JavaScript vs. TypeScript):** Although your preliminary stack mentioned TypeScript, you noted you are a beginner learning JavaScript, React, and JSX with under 24 hours. We will use **React + JavaScript (.jsx)** in Vite rather than TypeScript (.tsx). *Why:* Fighting TypeScript compiler errors on Supabase return types can easily burn 3–4 hours of a 24-hour hackathon without adding visible value for judges.
2. **Assumption A-02 (Deterministic Readiness Gates):** The Readiness Engine will evaluate **6 specific mandatory rules** (defined in Section A.3) covering Deal Metadata, Scope & Deliverables, Access/Dependency Checklist, Supporting Document Links, and Delivery Owner Assignment.
3. **Assumption A-03 (Demo Role Switcher over Login Screens):** Instead of email/password registration, the top navigation bar will feature an instant **Demo Role Switcher** with three pre-seeded personas (Sales Rep, Delivery Lead, Agency Ops Manager).
4. **Assumption A-04 (Document Links vs. Binary File Storage):** Supporting documents (Signed Proposal, Brand Assets, Technical Brief) will be stored as validated URLs (e.g.,

Notion, Figma, Google Drive links) rather than raw file uploads. *Why:* Configuring cloud storage buckets, CORS policies, and multipart file upload forms adds 2+ hours of risk when real agencies actually share cloud links anyway.

Deliverable A: Product Scope, MVP Boundaries & Readiness Rules

A.1 Problem Statement & Value Proposition

When a service agency closes a deal, vital context—scope boundaries, promised deliverables, client dependencies, deadlines, and document links—is fragmented across Slack threads, sales calls, and emails. Delivery teams inherit half-baked projects, causing scope creep, missed commitments, and damaged client trust.**HandoffHQ** replaces messy ad-hoc handoffs with a gated **Readiness Engine** that refuses to let Sales mark a project "Ready for Delivery" until every mandatory operational condition is satisfied and verified.

A.2 Strict MVP Boundaries (Under-24-Hour Scope Lock)

- **IN SCOPE (MVP):**
  - Agency Command Center dashboard with KPI summary cards (Total Value in Flight, Blocked Handovers, Ready for Delivery, Handed Over) and status filters.
  - Pre-seeded realistic handovers, headlined by **"Acme Website Redesign"** (starts in Blocked state with 3 specific missing items for the 60-second demo).
  - Interactive Handover Workspace with live editing of project overview, scope, structured deliverables, required access/dependencies, document links, and delivery owner.
  - Real-time deterministic **Readiness Engine** panel displaying a progress bar, exact blockers, "Why it blocks" explanations, and 1-click "Jump to fix" or inline resolution.
  - Enforced state machine (Draft / Blocked → Ready for Delivery → Handed Over) gated by role and readiness score.
  - Automatic **Activity History / Audit Log** recording every creation, blocker resolution, owner assignment, and final sign-off with timestamps and actor roles.
  - 1-Click **"Reset Demo Data"** button in the header so you can reset "Acme Website Redesign" back to its initial Blocked state in 2 seconds between judge visits.
- **OUT OF SCOPE (Cut to Protect the 24-Hour Deadline):**
  - Real OAuth / email-password login walls.
  - LLM/AI-generated summaries (explicitly excluded in favor of deterministic rules).
  - External webhooks (Slack/Email notifications, live HubSpot/Notion OAuth sync).
  - Binary file uploads (PDF/ZIP hosting).
  - Multi-tenant organization management.

A.3 The Deterministic Readiness Engine Rules (6 Mandatory Gates)

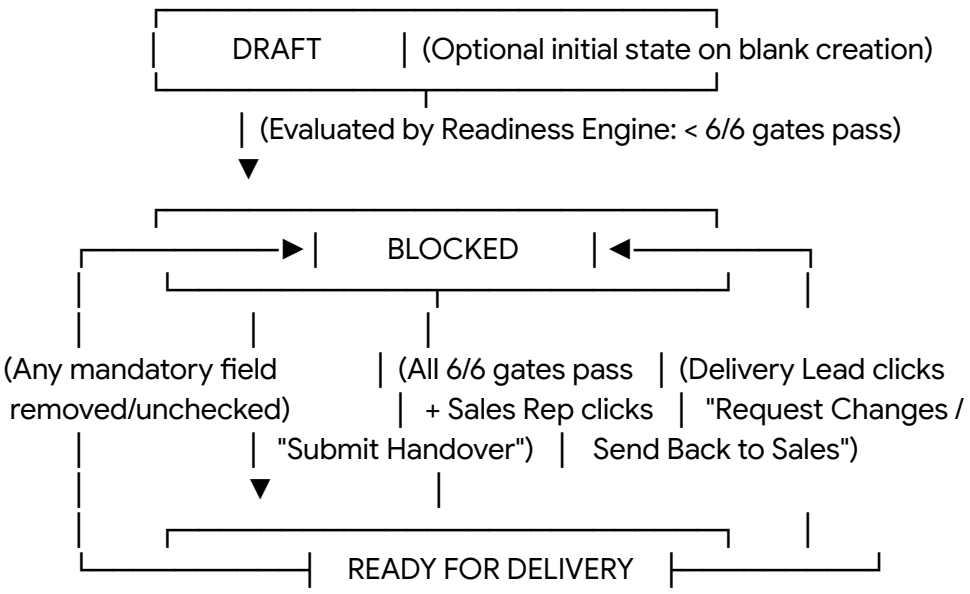
Every time a handover record or its child checklists change, the pure function

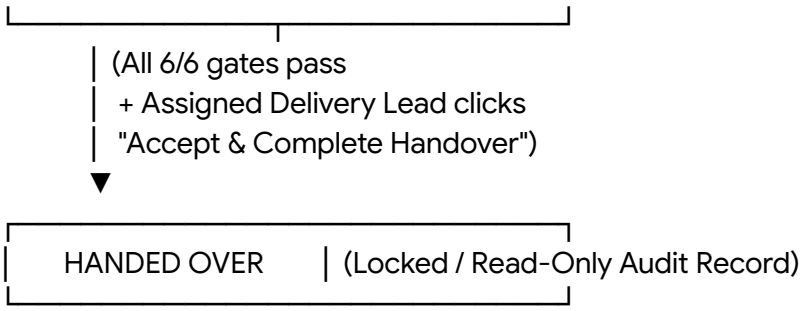
evaluateReadiness(handover) runs on the client and before any status transition request. All 6 gates must pass (100%) to unlock the **"Mark Ready for Delivery"** button:

Gate ID	Rule Name	Deterministic Pass Condition	"Why This Blocks Handover" Explanation	Action Needed to Resolve
R-01	**Commercial & Timeline Baseline**	client_name non-empty, deal_value > 0, and target_launch_date is set.	Delivery cannot schedule capacity or track budget without confirmed deal value and target launch date.	Fill in the Client Name, Deal Value ($), and Target Launch Date in Overview.
R-02	**Scope Summary**	scope_summary trimmed length ≥ 30 characters.	Ambiguous or one-word scope notes lead to immediate scope disputes with the client during kickoff.	Write a clear scope summary (at least 30 characters) outlining core project boundaries.
R-03	**Agreed Deliverables**	At least 2 items exist in deliverables array, each with a title and due date.	Delivery managers cannot build a sprint plan without itemized deliverables and milestone dates.	Add at least 2 concrete deliverables with target dates in the Deliverables section.
R-04	**Mandatory Client Dependencies**	Every item in dependencies where is_mandatory	Starting engineering or design work without client	Check off / verify all mandatory client access &

		=== true has is_resolved === true.	credentials/assets stalls the team mid-sprint.	dependency items.
R-05	**Supporting Documents**	At least 1 valid URL (https://...) in document_links (e.g., Signed SOW / Proposal).	Verbal sales promises are unenforceable without a linked SOW, proposal, or design spec.	Attach at least one valid https:// link to the signed proposal or spec document.
R-06	**Assigned Delivery Owner**	delivery_owner_id is non-null / non-empty.	An unowned handover sits in limbo with zero accountability for client kickoff.	Select and assign a receiving Delivery Lead from the owner dropdown.

A.4 Valid Status Transitions & State Machine





Deliverable B: User Personas & Core User Journeys

B.1 Personas (Pre-Seeded in Demo Role Switcher)

1. **Aarav Mehta — Senior Account Executive (Role: sales)**
   - **Goal:** Hand off newly closed deals quickly without getting pulled into delivery firefighting two weeks later.
   - **Permissions:** Can create handovers, edit fields/checklists on Draft/Blocked handovers, and click **"Mark Ready for Delivery"** once 6/6 gates pass. Cannot accept a handover on behalf of Delivery.
2. **Priya Nair — Delivery Lead / Technical PM (Role: delivery)**
   - **Goal:** Never inherit a project with missing credentials, vague scope, or unverified promises.
   - **Permissions:** Can inspect handovers, verify dependencies, click **"Accept & Complete Handover"** (when status is Ready for Delivery), or **"Send Back to Sales"** with a note.
3. **Vikram Desai — Head of Agency Operations (Role: ops)**
   - **Goal:** Bird's-eye visibility across all active transitions to spot revenue stuck in Blocked status.
   - **Permissions:** Can view all metrics, filter handovers by status, inspect audit trails, and trigger demo resets.

B.2 The 60-Second Judge Demo Journey ("Acme Website Redesign")

1. **0:00–0:15 (The Problem & Visibility):** Judge sees the Command Center dashboard as **Aarav (Sales Rep)**. **"Acme Website Redesign ($45,000)"** is highlighted in amber/red as **Blocked (3/6 Gates Passed — 50%)**.
2. **0:15–0:35 (The Readiness Engine in Action):** Judge clicks into "Acme Website Redesign". The sticky right-hand **Readiness Engine Panel** clearly lists 3 red blockers:
   - *Blocker 1 (R-02):* Scope Summary is too brief (*"Redesign site"* — 13 chars).
   - *Blocker 2 (R-04):* Mandatory dependency *"Production DNS & Webflow Workspace Access"* is unchecked.
   - *Blocker 3 (R-06):* Delivery Owner is unassigned.
   - Notice that the **"Mark Ready for Delivery"** button is disabled with a lock icon.

3. **0:35–0:48 (Live Resolution):** Judge clicks **"Quick Fill Scope"** (or types a detailed scope sentence), checks the box for *"Production DNS & Webflow Workspace Access"*, and selects **Priya Nair (Delivery Lead)** as the Delivery Owner. Instantly, the Readiness Engine hits **100% (6/6 Gates Passed)**, turns emerald green, and unlocks **"Mark Ready for Delivery"**. Judge clicks it; status shifts to **Ready for Delivery**.
4. **0:48–1:00 (Role Handoff & Audit Trail):** Judge switches the top navbar persona dropdown from **Aarav (Sales)** to **Priya (Delivery Lead)**. Priya sees the green banner inviting her to review and accept. Judge clicks **"Accept & Complete Handover"**. Status locks as **Handed Over**, confetti/success badge appears, and the **Activity Log** at the bottom shows every timestamped step.

Deliverable C: Prioritized Feature List & Acceptance Criteria

Must-Have Features (P0 — Required for Hackathon Demo)

ID	Feature	User Story	Acceptance Criteria (Testable)
F-01	**Handover Command Center Dashboard**	As Ops/Sales/Delivery, I want to see all handovers, their readiness %, and status KPIs at a glance.	1. Displays 4 KPI cards: Total Active Pipeline ($), Blocked Count, Ready for Delivery Count, Handed Over Count. 2. Lists handover cards with client name, project title, deal value, status badge, readiness progress bar (e.g., 4/6 Gates), and owner. 3. Includes status filter tabs (All, Blocked, Ready for Delivery, Handed Over).

F-02	**Deterministic Readiness Engine Panel**	As a Sales Rep, I want real-time feedback on what blocks my handover and why.	1. Evaluates Rules R-01 through R-06 immediately on every state change. 2. Displays overall score (X/6 Gates Passed, 0–100%). 3. For each failed rule, renders: Rule Title, "Why it blocks" explanation, and "Action needed" instruction. 4. Disables the "Mark Ready for Delivery" button whenever < 6 rules pass.
F-03	**Structured Handover Editor**	As a Sales Rep, I want to edit project details, deliverables, dependencies, links, and owner in one workspace.	1. Allows editing Overview (project_name, client_name, deal_value, target_launch_date, scope_summary, delivery_owner_id). 2. Allows adding/deleting Deliverables (title, due_date). 3. Allows toggling is_resolved on Dependencies and adding new dependencies (title, is_mandatory).

			4. Allows adding/removing Document Links (label, url validated for http:// or https://). 5. Persists all changes and logs activity entries.
F-04	**Role-Gated State Transitions**	As a Delivery Lead, I want to formally accept or reject a ready handover so accountability transfers cleanly.	1. Only sales (or ops) can click "Mark Ready for Delivery" when 6/6 gates pass. 2. Only delivery (or ops) can click "Accept & Complete Handover" or "Send Back to Sales" when status is Ready for Delivery. 3. Once Handed Over, all form inputs become read-only to preserve audit integrity.
F-05	**Immutable Activity & Audit Log**	As an Ops Manager, I want a chronological log of who changed what and when.	1. Every status change, owner assignment, and checklist resolution appends an entry to activity_logs. 2. Displays actor name, actor role

			badge, action description, and relative/formatted timestamp in reverse chronological order.
F-06	**1-Click Demo Seed & Reset**	As a presenter, I want to reset "Acme Website Redesign" to its exact starting demo state in one click.	1. Navbar includes a "Reset Demo Data" button. 2. Clicking it restores the 4 pre-seeded agency projects (with "Acme Website Redesign" at 3/6 gates in Blocked status) in under 2 seconds.

Should-Have Features (P1 — Build Only if Ahead of Schedule at Hour 15)

- **F-07: "Quick-Fix / Fill Sample Text" Helpers in Demo Mode:** Small helper buttons next to empty fields during the demo so judges don't even have to type a 30-character scope summary manually. *(High ROI, takes 15 minutes to add)*.
- **F-08: Create New Handover Modal with Agency Template:** Pre-populates standard mandatory agency dependencies (e.g., Signed SOW, Brand Guidelines, Staging Access) when creating a brand-new deal.

Future Features (P2 — Post-Hackathon Roadmap)

- HubSpot / Pipedrive webhook ingestion on Deal Won.
- Slack bot notifications to #delivery-handoffs.
- Automated Notion project workspace provisioning upon handover acceptance.

Deliverable D: Recommended Tech Stack, Reasons & Verified Costs

Primary Stack Selection

Layer	Choice	Beginner-Friendly Explanation & Rationale	Verified Free Tier Limits & Cost ($0.00)
Frontend Framework	**React 18 + Vite (JavaScript .jsx)**	Uses the exact JS/React/JSX skills you already have. Vite spins up in <1 second and avoids Next.js SSR hydration bugs.	**100% Free & Open Source** (MIT License).
Styling & Icons	**Tailwind CSS + lucide-react**	Utility classes let you build a high-contrast B2B SaaS UI (status pills, progress bars, split panels) without context-switching to CSS files.	**100% Free & Open Source** (MIT License).
Backend & Database	**Supabase (Managed PostgreSQL + @supabase/supabase-js)**	**Zero custom backend server to deploy.** The React app talks directly to Supabase PostgreSQL via its auto-generated REST client. Saves 4–6 hours of writing and hosting an Express API.	**Verified Free Tier ($0/mo, no credit card required):** Includes 2 free projects, 500 MB database storage, 5 GB bandwidth, 50,000 monthly active users. *(Note: Free projects pause after 7 days of inactivity, which does not affect a 24-hour hackathon).*
Fail-Safe Data Adapter	**Dual-Mode Storage Service (Supabase with automatic**	Wraps all data calls in a single src/services/handoverService.js file. If	**100% Free** (Built into browser APIs).

	localStorage fallback)	Supabase is configured in .env, it syncs with Postgres. If Wi-Fi drops during judging or .env is empty, it seamlessly falls back to browser localStorage seeded with the exact same demo data.	
Deployment	**Vercel (Hobby Plan)**	Connects directly to your GitHub repo and deploys your Vite React app globally in ~45 seconds on every git push.	**Verified Free Tier ($0/mo, no credit card required):** 100 GB bandwidth/month, automatic HTTPS, unlimited personal deployments.

Deliverable E: Architecture Diagram & Component Responsibilities

System Architecture Diagram (Mermaid)

Code snippet
flowchart TB
  subgraph Client["Browser Client (React + Vite + Tailwind CSS)"]
    Nav["Top Navbar\n(Role Switcher: Sales / Delivery / Ops\n+ 1-Click Demo Reset)"]
    Dash["Command Center View\n(KPI Cards, Status Filters, Project Grid)"]
    Work["Handover Workspace View\n(Left: Editable Sections | Right: Readiness Panel)"]
    Engine["Readiness Engine (src/lib/readinessEngine.js)\nPure Deterministic Rule Evaluator (Rules R-01 to R-06)"]
    Adapter["Data Access Service (src/services/handoverService.js)\nUnified Async CRUD

```
Interface"]
  end

  subgraph Persistence["Persistence Layer (Dual-Mode)"]
    Supa["Primary: Supabase Cloud\n(PostgreSQL Database + Auto REST API)"]
    Local["Fallback: Browser localStorage\n(Auto-activates if offline or no API key)"]
  end

  Nav --> Dash
  Nav --> Work
  Dash --> Adapter
  Work --> Engine
  Work --> Adapter
  Engine -->|Returns Score, Status, Blockers & Actions| Work
  Adapter -->|When VITE_SUPABASE_URL present| Supa
  Adapter -.->|Instant Offline Fallback| Local
```

Component Responsibilities

1. **App.jsx & Top Navbar (RoleContext):** Holds the currently active demo user (Aarav - Sales, Priya - Delivery, or Vikram - Ops), manages switching between the Dashboard view and the active Handover Workspace view, and triggers the 1-click Demo Reset.
2. **readinessEngine.js (Pure Domain Logic):** A zero-dependency JavaScript module that accepts a handover object and returns { passedCount, totalCount: 6, percentage, canMarkReady, computedStatus, gateResults: [...] }. Because it is a pure function (same input always produces the exact same output with no database calls), it runs instantaneously as the user types or checks boxes.
3. **handoverService.js (Data Access Layer):** Centralizes all reads and writes (getHandovers, getHandoverById, updateHandover, transitionStatus, resetDemoData). Components never import Supabase directly—they call handoverService.js.
4. **Supabase PostgreSQL:** Stores the persistent state of handovers and their child items so refreshes and judge tests reflect real database persistence.

Deliverable F: Database Schema (Fields, Types, Relationships & Constraints)

Crucial ArchitecturalSimplification for the 24-Hour Deadline

In a large enterprise system, you might create 6 normalized tables (users, handovers, deliverables, dependencies, documents, activity_logs) requiring multi-table foreign-key joins and complex transactions on every checkbox click. For a **solo beginner in under 24 hours**, we use **PostgreSQL JSONB columns** for the embedded checklists (deliverables, dependencies,

document_links, activity_logs) inside a single **handovers** table.

- **Why this is a massive win for you:** Fetching a handover is 1 query (SELECT * FROM handovers). Toggling a dependency checkbox + logging an activity item + updating readiness status happens in **one atomic UPDATE call**, eliminating race conditions, foreign-key cascade bugs, and multi-table join errors.

SQL Schema (supabase/schema.sql)

SQL

```
-- Enable UUID generation
CREATE EXTENSION IF NOT EXISTS "pgcrypto";

CREATE TABLE IF NOT EXISTS public.handovers (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_name TEXT NOT NULL CHECK (char_length(trim(project_name)) > 0),
  client_name TEXT NOT NULL CHECK (char_length(trim(client_name)) > 0),
  deal_value NUMERIC(12, 2) NOT NULL DEFAULT 0 CHECK (deal_value >= 0),
  target_launch_date DATE NULL,

  -- Status constrained strictly to valid state machine states
  status TEXT NOT NULL DEFAULT 'Blocked'
    CHECK (status IN ('Draft', 'Blocked', 'Ready for Delivery', 'Handed Over')),

  -- Scope & Ownership
  scope_summary TEXT NOT NULL DEFAULT '',
  sales_owner_name TEXT NOT NULL DEFAULT 'Aarav Mehta (Sales Rep)',
  delivery_owner_id TEXT NULL,      -- e.g., 'usr_priya' or NULL when unassigned
  delivery_owner_name TEXT NULL,    -- e.g., 'Priya Nair (Delivery Lead)'

  -- Structured Checklist Arrays stored as JSONB for atomic updates
  -- deliverables: [{ id, title, due_date }]
  deliverables JSONB NOT NULL DEFAULT '[]'::jsonb,

  -- dependencies: [{ id, title, category, is_mandatory, is_resolved }]
  dependencies JSONB NOT NULL DEFAULT '[]'::jsonb,

  -- document_links: [{ id, label, url }]
  document_links JSONB NOT NULL DEFAULT '[]'::jsonb,

  -- activity_logs: [{ id, actor_name, actor_role, action, timestamp }]
  activity_logs JSONB NOT NULL DEFAULT '[]'::jsonb,
```

```
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- Index for fast dashboard filtering by status and sorting by updated_at
CREATE INDEX IF NOT EXISTS idx_handovers_status_updated
ON public.handovers (status, updated_at DESC);

-- Permissive RLS policy for hackathon demo (allows anon key read/write on demo table)
ALTER TABLE public.handovers ENABLE ROW LEVEL SECURITY;

CREATE POLICY "Allow public demo access to handovers"
ON public.handovers
FOR ALL
TO anon, authenticated
USING (true)
WITH CHECK (true);
```

Embedded JSONB Object Shapes

- **DeliverableItem**: { id: string, title: string, due_date: string (YYYY-MM-DD) }
- **DependencyItem**: { id: string, title: string, is_mandatory: boolean, is_resolved: boolean }
- **DocumentLinkItem**: { id: string, label: string, url: string }
- **ActivityLogItem**: { id: string, actor_name: string, actor_role: 'sales' | 'delivery' | 'ops', action: string, timestamp: string (ISO-8601) }

Deliverable G: API & Data-Access Contract

Because we use Supabase's auto-generated REST API wrapped inside src/services/handoverService.js, your React components interact with clean JavaScript async functions that map 1:1 to Supabase REST (PostgREST) calls.

G.1 Endpoint & Service Method Contract

Service Function	Underlying Supabase / HTTP Call	Auth / Role Requirement	Request Payload	Success Response (200/201)	Validation & Error Handling
listHandovers()	GET /rest/v1/handovers?select=*&order	Any role (sales, delivery, ops)	None	Array<HandoverWithReadiness>	If Supabase unreachable, switches to

	=updated_at.desc				localStorage and shows subtle "Offline Demo Mode" pill.
getHandover(id)	GET /rest/v1/handovers?id=eq.:id&select=*	Any role	id: string	HandoverWithReadiness	Returns 404 Not Found error banner if ID does not exist.
createHandover(data, actor)	POST /rest/v1/handovers	sales or ops	{ project_name, client_name, deal_value, target_launch_date }	Newly created Handover (seeded with default mandatory agency dependencies & initial activity log).	Rejects if project_name or client_name is empty (400 Validation Error).
saveHandoverChanges(id, updates, logMessage, actor)	PATCH /rest/v1/handovers?id=eq.:id	sales or ops (Blocked/Draft); locked if Handed Over	Partial fields (scope_summary, deliverables, dependencies, document_links, delivery_owner_id, etc.)	Updated HandoverWithReadiness	1. Blocks edit if current status is Handed Over. 2. Re-runs evaluateReadiness(). If status was Ready for Delivery and a required

					gate is now broken, auto-reverts status to Blocked.
markReadyForDelivery(id, actor)	PATCH /rest/v1/handovers?id=eq.:id	sales or ops	{ id, actor }	Updated Handover with status: 'Ready for Delivery' + audit log entry.	**Hard Gate:** Runs evaluateReadiness(handover). If passedCount < 6, throws 422 Unprocessable: Cannot mark ready while blockers remain.
acceptHandover(id, actor, note)	PATCH /rest/v1/handovers?id=eq.:id	delivery or ops	{ id, actor, note }	Updated Handover with status: 'Handed Over' + audit log entry.	**Hard Gate:** Verifies role is delivery/ops, current status is Ready for Delivery, and all 6 gates still pass.
returnToSales(id, actor, reason)	PATCH /rest/v1/handovers?id=eq.:id	delivery or ops	{ id, actor, reason }	Updated Handover with status: 'Blocked' + audit log entry explaining rejection	Requires non-empty reason string so Sales knows what to fix.

				reason.	
resetDemoData()	DELETE all + POST /rest/v1/handovers (bulk seed)	Any role (Demo Toolbar)	None	Fresh array of 4 seeded handovers	Restores "Acme Website Redesign" to exact Blocked (3/6) demo state.

Deliverable H: Authentication, Security & Error-Handling Plan

H.1 Frictionless Demo Auth (Role Switcher Pattern)

- **Why no login screen:** Judges evaluate 20+ projects in a row. Forcing a judge to sign up, wait for a confirmation email, and log out/log back in to test both the Sales Sender and Delivery Receiver views kills momentum.
- **How Role Switching Works:**
  - Three pre-configured user objects live in src/constants/demoUsers.js:
    1. { id: 'usr_aarav', name: 'Aarav Mehta', role: 'sales', title: 'Account Executive' }
    2. { id: 'usr_priya', name: 'Priya Nair', role: 'delivery', title: 'Delivery Lead' }
    3. { id: 'usr_vikram', name: 'Vikram Desai', role: 'ops', title: 'Head of Agency Ops' }
  - Both the UI buttons **and** the service layer check activeUser.role before permitting state transitions.
  - When activeUser.role === 'sales', the **"Accept & Complete Handover"** button displays a clear helper tooltip: *"Switch role to Priya Nair (Delivery Lead) in the top bar to accept this handover."* This guides the judge through the exact multi-role story effortlessly.

H.2 Input Validation & Security Essentials

1. **URL Validation on Document Links:** Any link added to document_links is validated with new URL(input) and checked for http: or https: protocol to prevent javascript: XSS injection and broken links.
2. **Environment Variable Safety:** Only VITE_SUPABASE_URL and VITE_SUPABASE_ANON_KEY are placed in client .env. Never commit .env to Git; commit .env.example instead.
3. **Graceful Error & Offline Handling:** Every async action wraps calls in try/catch, sets a button loading state (isSaving) to prevent double-clicks, displays a clean toast

notification on success/error, and falls back to localStorage if Supabase network requests fail.

Deliverable I: Project Folder Structure

Plaintext

```
handoffhq/
├── .env.example              # Template for VITE_SUPABASE_URL and VITE_SUPABASE_ANON_KEY
├── .gitignore
├── index.html
├── package.json
├── postcss.config.js
├── tailwind.config.js
├── vite.config.js
├── supabase/
│   └── schema.sql            # Copy-paste SQL script to create table & RLS in Supabase SQL Editor
└── src/
    ├── main.jsx              # React entry point
    ├── App.jsx               # Root layout, Role state, Active View router, Toast banner
    ├── index.css             # Tailwind directives (@tailwind base; @tailwind components; @tailwind utilities;)
    ├── constants/
    │   ├── demoUsers.js      # Pre-seeded personas (Sales, Delivery, Ops) & available delivery owners
    │   └── seedData.js       # 4 realistic agency handovers including "Acme Website Redesign"
    ├── lib/
    │   ├── supabaseClient.js # Initializes Supabase client if env vars exist, else null
    │   └── readinessEngine.js # PURE FUNCTION: Evaluates Rules R-01 to R-06 & returns blockers/status
    ├── services/
    │   └── handoverService.js # Unified CRUD + State Transitions (Supabase + localStorage fallback)
    └── components/
        ├── Navbar.jsx        # Brand header, Persona Role Switcher, "Reset Demo Data" button
        ├── DashboardView.jsx # 4 KPI cards, Status filter pills, Handover cards grid, "+ New Handover" modal
        ├── HandoverWorkspace.jsx # Split-screen workspace container (Left: Editor, Right: Readiness Engine)
```

├── ReadinessPanel.jsx # Sticky right sidebar: Score gauge, 6 Gate cards, Blocker explanations, Action CTA
├── DeliverablesSection.jsx # Add/remove deliverables with target dates (Rule R-03)
├── DependenciesSection.jsx # Toggle mandatory/optional access & client dependencies (Rule R-04)
├── DocumentsSection.jsx # Add/remove validated https:// document links (Rule R-05)
└── ActivityTimeline.jsx # Chronological audit log with actor role badges

Deliverable J: Ordered Implementation Roadmap (18-Hour Budget + 6-Hour Buffer)

Phase	Task ID & Description	Depends On	Est. Time	Completion Criteria (Definition of Done)
Phase 1: Foundation (2.0 hrs)	**T-01:** Scaffold Vite + React (JS) app, install tailwindcss, lucide-react, @supabase/supabase-js.	None	45m	npm run dev renders a styled Tailwind test badge with a Lucide icon.
	T-02: Create constants/demoUsers.js and constants/seedData.js (with "Acme Website Redesign" pre-configured with 3 specific blockers).	T-01	45m	Seed data exports 4 valid handover objects covering Blocked, Ready for Delivery, and Handed Over states.
	T-03: Set up Supabase project & run supabase/schema.sql (or skip	T-01	30m	handovers table exists in Supabase OR local adapter initializes

	to local mode first and connect in Phase 4).			cleanly.
Phase 2: Core Engine & Data Layer (2.5 hrs)	**T-04:** Implement src/lib/readinessEngine.js with Rules R-01 to R-06, blocker explanations, and remediation strings.	T-02	1.0h	Passing "Acme Website Redesign" returns 3/6 passed (50%), canMarkReady: false, and exact details for failing rules R-02, R-04, R-06.
	T-05: Implement src/services/handoverService.js with Dual-Mode persistence (Supabase + localStorage fallback) and state transition guards.	T-03, T-04	1.5h	Service functions listHandovers, saveHandoverChanges, markReadyForDelivery, acceptHandover, resetDemoData work and persist across browser refreshes.
Phase 3: Core UI & Demo Flow (5.5 hrs)	**T-06:** Build Navbar.jsx with the instant Persona Role Switcher (Sales, Delivery, Ops) and Reset Demo Data	T-02, T-05	1.0h	Switching persona updates global active user state; clicking Reset restores initial seed data.

	button.			
	T-07: Build DashboardView.jsx with 4 KPI summary cards, status filter tabs, handover cards with readiness progress bars, and a basic "+ New Handover" modal.	T-05, T-06	1.5h	Clicking "Acme Website Redesign" opens its detail workspace. KPI counts match actual handover statuses.
	T-08: Build ReadinessPanel.jsx (the star differentiator UI) displaying the live readiness bar, each of the 6 rules (green check vs. red blocker + why it blocks + action needed), and role-aware transition buttons.	T-04	1.5h	Failed gates stand out visually in amber/red with clear explanations; transition buttons enforce role & readiness rules.
	T-09: Build HandoverWorkspace.jsx + child editor sections (Overview,	T-05, T-08	1.5h	**WORKING DEMO CHECKPOINT (Hour 10):** You can complete the entire

	DeliverablesSection, DependenciesSection, DocumentsSection, ActivityTimeline). Include a "Quick Fill Sample Scope" helper button for rapid demoing.			60-second "Acme Website Redesign" demo end-to-end locally!
Phase 4: Polish, Deploy & Pitch Prep (4.0 hrs)	**T-10:** Visual polish: empty states, read-only lock banner when Handed Over, toast feedback notifications, responsive layout check.	T-09	1.5h	UI looks crisp, high-contrast, and executive-ready on a 13"–16" laptop screen.
	T-11: Push to GitHub, connect to Vercel, configure VITE_SUPABASE_URL & VITE_SUPABASE_ANON_KEY, and verify live production URL.	T-10	1.0h	Production .vercel.app URL loads, persists edits, and resets cleanly.
	T-12: Rehearse the 60-second "Acme Website	T-11	1.5h	Demo flows smoothly in <60 seconds

	Redesign" judge demo 5 times and record a 90-second backup screen-captur e video.			with zero typing fumbles.

Deliverable K: Risks, Assumptions & Fallback Options

Risk / Failure Scenario	Likelihood	Impact	Concrete Mitigation & Instant Fallback Plan
R-1: Supabase setup blocks progress or venue Wi-Fi drops during live judging	Medium	High	**Built-in Dual-Mode Service:** handoverService.js automatically checks if Supabase is reachable. If env vars are missing or a network request fails, it transparently reads/writes to browser localStorage pre-seeded with seedData.js. The app never crashes or shows a blank screen.
R-2: Judge tests the app right after another judge already completed "Acme Website	High	High	**1-Click "Reset Demo Data" Button:** Placed permanently in the top navigation bar.

Redesign"			One click resets both Supabase and localStorage back to the exact starting state in <1 second.
R-3: Typing a 30+ character scope summary live during a 60-second pitch wastes 15 seconds	High	Medium	**"Insert Sample Scope" Quick-Action Pill:** Include a small button next to the Scope field in the UI that populates a realistic 2-sentence scope summary for Acme Corp in one click.
R-4: Vercel build fails at the 11th hour due to strict TypeScript/linter rules	Medium	High	Use plain **JavaScript (.jsx)** instead of TypeScript, and run npm run build locally at Hour 10 to verify zero build errors early.

Decision Log & Items for Your Review

ID	Decision	Rationale	Status
D-01	Use **React + Vite (JavaScript .jsx)** instead of TypeScript	Matches your existing JS/React/JSX skillset and prevents 24-hour hackathon delays caused by complex TypeScript	**Proposed (Recommended)**

		generics.	
D-02	Single handovers table in Supabase Postgres with JSONB columns for checklists & logs	Eliminates 5-table join queries and foreign-key transaction bugs while keeping full relational querying on status and deal value.	**Proposed (Recommended)**
D-03	Dual-Mode Data Service (Supabase primary + automatic localStorage fallback)	Guarantees a working demo even before you configure Supabase or if hackathon venue Wi-Fi drops.	**Proposed (Recommended)**
D-04	Top-bar **Persona Role Switcher** (Sales, Delivery, Ops) instead of email/password login	Lets judges test the two-sided Sales-to-Delivery handoff in 60 seconds without logging out and back in.	**Approved (Per Brief)**
D-05	6 Deterministic Readiness Gates (Rules R-01 to R-06)	Predictable, explainable, fast to test, and directly highlights the "Acme Website Redesign" demo blockers.	**Approved (Per Brief)**

Deliverable L: Self-Contained Coding-Agent Handoff Specification

Instructions for User: Once you approve this blueprint (or tell me to write the code directly), copy and paste the entire block below into your Coding Agent (or ask me to execute it step-by-step). It contains every requirement, rule, data structure, and file path needed to build

the complete HandoffHQ MVP without external context.

Markdown
CODING AGENT SPECIFICATION: HandoffHQ MVP

1. Project Overview
Build **HandoffHQ**, a Sales-to-Delivery Handover Command Center for service agencies. Its core differentiator is a deterministic, rule-based **Readiness Engine** that evaluates whether a newly won client project has satisfied all 6 mandatory operational gates before allowing a Sales Rep to mark it `Ready for Delivery`, and before allowing a Delivery Lead to formally accept it as `Handed Over`.

2. Tech Stack & Constraints
- **Framework:** React 18 + Vite using plain JavaScript (`.jsx`) — DO NOT use TypeScript.
- **Styling:** Tailwind CSS + `lucide-react` icons.
- **Persistence:** `@supabase/supabase-js` connected to a single `handovers` PostgreSQL table, WITH an automatic `localStorage` fallback inside `src/services/handoverService.js` so the app works immediately out-of-the-box with zero configuration and survives offline conditions.
- **Authentication:** Top-navbar **Demo Role Switcher** allowing instant toggling between 3 personas:
  1. `Aarav Mehta` (`role: 'sales'`, Account Executive)
  2. `Priya Nair` (`role: 'delivery'`, Delivery Lead)
  3. `Vikram Desai` (`role: 'ops'`, Head of Agency Ops)

3. Deterministic Readiness Engine Specification (`src/lib/readinessEngine.js`)
Create a pure function `evaluateReadiness(handover)` that evaluates these exact 6 mandatory rules in order:

1. **R-01: Commercial & Timeline Baseline**
   - **Pass Condition:** `Boolean(handover.client_name?.trim()) && Number(handover.deal_value) > 0 && Boolean(handover.target_launch_date)`
   - **Why it blocks:** "Delivery cannot forecast capacity or budget without a confirmed client, deal value, and target launch date."
   - **Action needed:** "Enter a positive Deal Value and Target Launch Date in Project Overview."

2. **R-02: Detailed Scope Summary**
   - **Pass Condition:** `(handover.scope_summary?.trim().length || 0) >= 30`
   - **Why it blocks:** "Vague or one-line scope notes cause immediate scope creep and client disputes during kickoff."
   - **Action needed:** "Write a clear Scope Summary of at least 30 characters defining project boundaries."

3. **R-03: Itemized Deliverables**
   - **Pass Condition:** `Array.isArray(handover.deliverables) && handover.deliverables.length >= 2 && handover.deliverables.every(d => d.title?.trim() && d.due_date)`
   - **Why it blocks:** "Delivery leads cannot construct a sprint schedule without at least 2 concrete deliverables and target dates."
   - **Action needed:** "Add at least 2 deliverables with valid milestone dates."

4. **R-04: Mandatory Access & Client Dependencies**
   - **Pass Condition:** `Array.isArray(handover.dependencies) && handover.dependencies.filter(d => d.is_mandatory).length >= 1 && handover.dependencies.filter(d => d.is_mandatory).every(d => d.is_resolved === true)`
   - **Why it blocks:** "Starting execution without mandatory client credentials or environment access stalls the engineering team mid-sprint."
   - **Action needed:** "Verify and check off all mandatory client dependencies."

5. **R-05: Supporting Documentation Linked**
   - **Pass Condition:** `Array.isArray(handover.document_links) && handover.document_links.length >= 1 && handover.document_links.every(doc => doc.label?.trim() && /^https?:\/\/.+/.test(doc.url?.trim()))`
   - **Why it blocks:** "Verbal sales commitments are unenforceable without a linked SOW, signed proposal, or technical specification."
   - **Action needed:** "Attach at least one valid https:// link to supporting documentation."

6. **R-06: Assigned Receiving Delivery Owner**
   - **Pass Condition:** `Boolean(handover.delivery_owner_id && handover.delivery_owner_name)`
   - **Why it blocks:** "An unassigned handover has zero accountability for client onboarding and kickoff."
   - **Action needed:** "Assign a receiving Delivery Lead from the Delivery Owner dropdown."

Return Signature of `evaluateReadiness(handover)`:
```js
{
  passedCount,      // 0 to 6
  totalCount: 6,
  percentage,       // Math.round((passedCount / 6) * 100)
  isReady: passedCount === 6,
  gates: [          // All 6 rules with { id, title, passed, whyItBlocks, actionNeeded }
  ],
  blockers: [       // Filtered array of gates where passed === false
  ]
}
```

4. State Machine & Role Authorization Rules

- Valid statuses: 'Draft', 'Blocked', 'Ready for Delivery', 'Handed Over'.
- While editing a handover that is not 'Handed Over':
  - If status was 'Ready for Delivery' and an edit causes passedCount < 6, automatically downgrade status to 'Blocked' and log an activity entry.
- **Transition 1: Mark Ready for Delivery (Blocked/Draft -> Ready for Delivery)**
  - Requires: evaluateReadiness(handover).isReady === true AND activeUser.role in ['sales', 'ops'].
  - Appends activity log: "Marked handover Ready for Delivery (6/6 readiness gates passed)".
- **Transition 2: Accept & Complete Handover (Ready for Delivery -> Handed Over)**
  - Requires: handover.status === 'Ready for Delivery' AND evaluateReadiness(handover).isReady === true AND activeUser.role in ['delivery', 'ops'].
  - Appends activity log: "Accepted handover into Delivery execution. Transition marked Handed Over.".
  - Locks all inputs in the workspace (read-only audit mode).
- **Transition 3: Send Back to Sales (Ready for Delivery -> Blocked)**
  - Requires: handover.status === 'Ready for Delivery' AND activeUser.role in ['delivery', 'ops'].
  - Appends activity log with the reason provided.

5. Pre-Seeded Demo Data (src/constants/seedData.js)

Seed 4 realistic agency handovers:

1. **"Acme Website Redesign"** (Client: Acme Corp, Value: $45,000, Status: 'Blocked'):
   - **Must start with 3 passed gates (R-01, R-03, R-05) and 3 failing gates (R-02, R-04, R-06)** for the 60-second judge demo:
     - scope_summary: "Redesign site" (13 chars — fails R-02). Include a 1-click UI helper button "✨ Insert Full Scope Summary" so the presenter can fix R-02 in 1 second during the live demo.
     - deliverables: 2 valid items ("Responsive Figma Design System", "Webflow CMS Migration & QA" — passes R-03).
     - dependencies: 2 mandatory items: "Brand Guidelines & Logo Kit" (is_resolved: true) and "Production DNS & Webflow Workspace Access" (is_resolved: false — fails R-04).
     - document_links: 1 valid link ("Signed Master Services Agreement (SOW)" -> "https://notion.so/acme-sow-2026" — passes R-05).
     - delivery_owner_id: null, delivery_owner_name: null (fails R-06).
2. **"FinPulse Mobile App MVP"** (Client: FinPulse Technologies, Value: $82,000, Status: 'Ready for Delivery', 6/6 gates passed, assigned to Priya Nair).

3. **"Nexus B2B SEO & Content Retainer"** (Client: Nexus Cloud, Value: $18,500, Status: 'Handed Over', 6/6 gates passed, assigned to Priya Nair).
4. **"VoltEdge Shopify Plus Migration"** (Client: VoltEdge Gear, Value: $34,000, Status: 'Blocked', 2/6 gates passed).

6. Required Views & Components

1. **Top Navbar (Navbar.jsx):**
    - App logo + "HandoffHQ — Sales-to-Delivery Command Center".
    - Storage mode badge ("Supabase Connected" or "Local Demo Mode").
    - **Persona Switcher Dropdown/Pills:** Instant switch between Aarav (Sales), Priya (Delivery), and Vikram (Ops) with distinct role color badges.
    - **"Reset Demo Data" Button:** Restores all 4 seeded projects in 1 click.
2. **Command Center Dashboard (DashboardView.jsx):**
    - 4 KPI Cards: Total Pipeline Value ($), Blocked Handovers (Red/Amber), Ready for Delivery (Blue/Indigo), Handed Over (Emerald).
    - Filter Tabs (All, Blocked, Ready for Delivery, Handed Over) + Search input.
    - Grid/List of Handover Cards showing Client, Project Name, Deal Value, Target Launch Date, Delivery Owner, Readiness Progress Bar (X/6 Gates), Status Badge, and "Open Command Workspace ->" button.
    - "+ New Handover" button & modal.
3. **Handover Workspace (HandoverWorkspace.jsx + ReadinessPanel.jsx):**
    - **Left Column (7 cols):**
        - Header with Back button, Status badge, Client & Deal metadata.
        - Section 1: Project Overview & Scope Summary (with character counter for Rule R-02, "✨ Insert Full Scope Summary" quick-fill button, and Delivery Owner selector for Rule R-06).
        - Section 2: Agreed Deliverables list + Add Deliverable form (Rule R-03).
        - Section 3: Client Access & Dependencies checklist with Mandatory tags and checkboxes (Rule R-04).
        - Section 4: Supporting Document Links with URL validation (Rule R-05).
        - Section 5: Chronological Activity & Audit Log (ActivityTimeline.jsx).
    - **Right Column (5 cols, sticky): ReadinessPanel.jsx:**
        - Large Readiness Gauge (X/6 Mandatory Gates Passed + percentage bar).
        - Expandable/visible list of all 6 gates: failed gates highlighted in rose/amber showing **Why it blocks** and **Action needed**; passed gates shown in emerald.
        - Role-aware Primary Action Box at the top/bottom of the panel:
            - If < 6 gates pass: Disabled button "🔒 Resolve X Blocker(s) to Mark Ready".
            - If 6/6 gates pass & status is Blocked: Enabled button "🚀 Mark Ready for Delivery" (for Sales/Ops).
            - If status is Ready for Delivery: Shows "✅ Accept & Complete Handover" and "↩️ Send Back to Sales" buttons (enabled for

Delivery/Ops; if current user is Sales, shows a helpful prompt to switch persona to Priya in the navbar).

- If status is Handed Over: Shows an emerald "🏆 Handover Complete & Locked" audit badge.

Next Actionable Task & How to Proceed

1. **Review the Decision Log (D-01 through D-05) above.**
2. **Your Next Step:**
   * If you want **me** to build the application code now, simply reply: **"Approved. Generate the complete application code."**
   * Or, if you are using a separate coding agent (e.g., Cursor, Windsurf, Copilot, or Claude Code), copy **Deliverable L** above and paste it directly as its system/initial prompt.